

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2022-2023
FULL MARKS: 75

Math 4641: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Explain how numerical approaches differ from analytical approaches to solving a problem.

4
(CO4)
(PO1)

Solution:

Any 2 valid differences will suffice. Some differences can be written as follows —

Criteria	Numerical Methods	Analytical Methods
Type of yielded solution	Approximate solution	Exact solution
Working Procedure	Keep iterating until the obtained solution is a “good enough” approximation.	Perform symbolic manipulations to arrive at an equation that yields the exact solution.
Example	Solving $ax^2 + bx + c = 0$ using the quadratic equation $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Solving $ax^2 + bx + c = 0$ using any root-finding algorithm e.g. the Bisection method.

Rubric:

- 2 points for each valid difference.

- b) Derive the formula for the Secant method geometrically.

6
(CO4)
(PO1)

Solution:

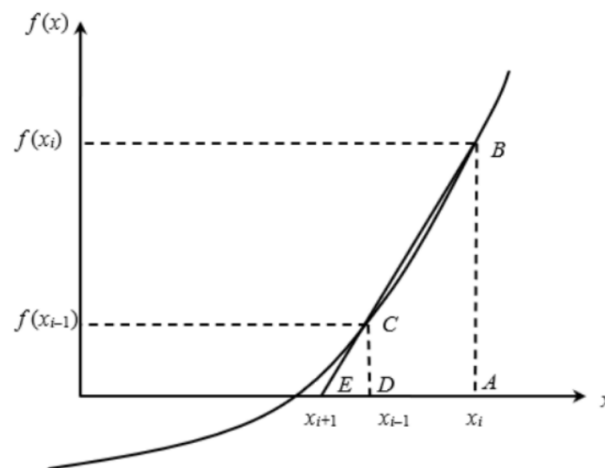


Figure 1: Geometrical representation of the Secant method.

Here, ABE and DCE are similar triangles.

Hence,

$$\begin{aligned}
 \frac{AB}{AE} &= \frac{DC}{DE} \\
 \frac{f(x_i)}{x_i - x_{i+1}} &= \frac{f(x_{i-1})}{x_{i-1} - x_{i+1}} \\
 \Rightarrow f(x_i)(x_{i-1} - x_{i+1}) &= f(x_{i-1})(x_i - x_{i+1}) \\
 \Rightarrow f(x_i)x_{i-1} - x_{i+1}f(x_i) &= x_if(x_{i-1}) - x_{i+1}f(x_{i-1}) \\
 \Rightarrow x_{i+1}f(x_{i-1}) - x_{i+1}f(x_i) &= f(x_{i-1})x_i - f(x_i)x_{i-1} \\
 \Rightarrow x_{i+1} &= \frac{x_if(x_{i-1}) - f(x_i)x_{i-1}}{f(x_{i-1}) - f(x_i)} + x_i - x_i \\
 \Rightarrow x_{i+1} &= x_i + \frac{x_if(x_{i-1}) - f(x_i)x_{i-1} - x_if(x_{i-1}) + x_if(x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i + \frac{-f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}
 \end{aligned}$$

Hence, we get the formula for the Secant method,

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Rubric:

- 4 points for a correct geometric derivation.
- 2 points for the correct formula.

- c) Figure 2 portrays the orbital geometry of planet Earth around the Sun, where S denotes the position of the Sun and E denotes the position of planet Earth.

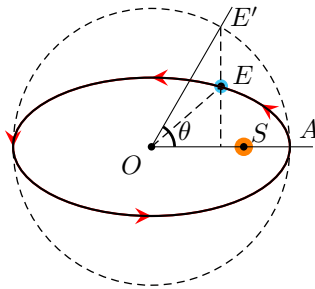


Figure 2: Geometry of the planetary orbit of Earth around the Sun for Question 1.c

Let θ denote the angle defined by $\angle E'OA$, measured in radians. The dotted circle is concentric to the ellipse and has a diameter equal to the major axis of the ellipse. Let T be the total period of planet Earth, and let t be the time it takes for Earth to go from A to E . The relationship between θ and t is dictated by Kepler's equation from orbital mechanics,

$$\theta - \epsilon \sin(\theta) = \frac{2\pi t}{T} \quad (1)$$

Here ϵ is the eccentricity of the elliptical orbit. In the case of planet Earth, the conventional value is $\epsilon = 0.017$.

- i. Formulate a function $f(\theta)$ such that its root will yield the value of the angle θ for a given value of t .

2.5
(CO1)
(PO2)

Solution:

The given equation is,

$$\theta - \epsilon \sin(\theta) = \frac{2\pi t}{T}$$

For planet Earth, the value of ϵ is 0.017, so the equation becomes,

$$\begin{aligned}\theta - 0.017 \sin(\theta) &= \frac{2\pi t}{T} \\ \Rightarrow \theta - 0.017 \sin(\theta) - \frac{2\pi t}{T} &= 0\end{aligned}$$

So the function $f(\theta)$ whose root yields the value of the angle θ for a given value of t is,

$$f(\theta) = \theta - 0.017 \sin(\theta) - \frac{2\pi t}{T}$$

Rubric:

- 2.5 points for the correct function $f(\theta)$.

- ii. Apply the Newton-Raphson method to estimate the value of θ corresponding to $t = \frac{T}{8}$. The initial guess is $\theta_0 = \pi$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table 1 and fill it out after performing the necessary calculations.

11.5
(CO2)
(PO1)

Table 1: The relevant values obtained in the answer of Question 1.c

Iteration	θ_i	$f(\theta_i)$	$f'(\theta_i)$	θ_{i+1}	$ \epsilon_a \%$	m
1	π					
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	θ_i	$f(\theta_i)$	$f'(\theta_i)$	θ_{i+1}	$ \epsilon_a \%$	m
1	π	$\frac{3\pi}{4}$	1.017	0.824783912	280.8988764%	0
2	0.824783912	0.02690092927	0.9884618335	0.7975689722	3.412236524%	1
3	0.7975689722	$4.58439207 \times 10^{-6}$	0.9881263745	0.7975643327	$5.817059919 \times 10^{-4}$	4
4	0.7975643327	-2.0113×10^{-11}	0.988126318	0.7975643327	$2.552019839 \times 10^{-9}$	10

Note: The answers delineated in the table above are highly precise and aren't indicative of other validly approximated answers.

Rubric:

- 0.5 point for each approximately correct answer.
- 1 point for demonstrating the step-by-step procedure in each iteration.

2. a) Define Truncation Error. What are the different avenues through which Truncation Error can be introduced? Explain with proper examples.

5
(CO4)
(PO1)

Solution:

Truncation Error is the error caused by approximating the mathematical process or the modeling technique. Some of the ways it can be introduced are —

1. **Chopping off terms in a series:** If we consider only the first 3 terms of the Maclaurin series of e^x , we get $e^x \approx \left(1 + x + \frac{x^2}{2!}\right)$. And so the truncation error is,

$$\begin{aligned}\text{Truncation Error} &= \text{Exact Value} - \text{Approximate Value} \\ &= e^x - \left(1 + x + \frac{x^2}{2!}\right)\end{aligned}$$

2. **Using a finite step size instead of an infinitesimal value:** To find the derivative of a function $f(x)$ using a finite step size h instead of an infinitesimally small h , the truncation error is,

$$\begin{aligned}\text{Truncation Error} &= \text{Exact Value} - \text{Approximate Value} \\ &= \left(\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}\right) - \left(\frac{f(x+h) - f(x)}{h}\right)\end{aligned}$$

3. **Approximating area under curve with rectangles:** To find the integral value of a function $f(x)$ within given bounds a and b using n rectangles of width Δx_i and height $f(x_i)$, the truncation error is,

$$\begin{aligned}\text{Truncation Error} &= \text{Exact Value} - \text{Approximate Value} \\ &= \int_a^b f(x)dx - \sum_{i=0}^n f(x_i) \cdot \Delta x_i\end{aligned}$$

Rubric:

- 1 point for a correct definition.
- 2 points for each valid example. Any 2 valid examples will suffice.

- b) The Taylor series for a function $f(x)$ is —

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

18
(CO4)
(PO1)

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of —

- i. $\ln(1+x)$
- ii. $\ln(1-x)$
- iii. $\ln\left(\frac{1+x}{1-x}\right)$
- iv. $\log_{(1-x)}(1+x)$

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$, and $\frac{d}{dx}(x^n) = nx^{n-1}$.

- (i) Now, for $f(x) = \ln(1+x)$, let's calculate the derivatives at the point $x = 0$.

$$\begin{aligned}f(x) &= \ln(1+x), f(0) = \ln(1+0) = \ln(1) = 0; f'(x) = \frac{1}{1+x}, f'(0) = \frac{1}{1+0} = 1; \\ f''(x) &= -\frac{1}{(1+x)^2}, f''(0) = -\frac{1}{(1+0)^2} = -1; f'''(x) = \frac{2}{(1+x)^3}, f'''(0) = \frac{2}{(1+0)^3} = 2; \\ f^{(4)}(x) &= -\frac{6}{(1+x)^4}, f^{(4)}(0) = -\frac{6}{(1+0)^4} = -6;\end{aligned}$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned} f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots \\ \Rightarrow f(h) &= 0 + h - \frac{h^2}{2!} + \frac{2h^3}{3!} - \frac{6h^4}{4!} + \dots \\ \Rightarrow f(x) &= 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \\ \Rightarrow \ln(1+x) &= 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \\ \Rightarrow \ln(1+x) &= x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \end{aligned}$$

(ii) **Short version:** Here,

$$\begin{aligned} f(x) &= \ln(1-x) \\ &= \ln(1+(-x)) \\ &= (-x) - \frac{(-x)^2}{2!} + \frac{2(-x)^3}{3!} - \frac{6(-x)^4}{4!} + \dots \quad [\text{using the Maclaurin series of } \ln(1+x) \text{ from (i)}] \\ &= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \end{aligned}$$

Long version: Now, for $f(x) = \ln(1-x)$, let's calculate the derivatives at the point $x = 0$.

$$\begin{aligned} f(x) &= \ln(1-x), f(0) = \ln(1-0) = \ln(1) = 0; f'(x) = -\frac{1}{1-x}, f'(0) = -\frac{1}{1-0} = -1; \\ f''(x) &= -\frac{1}{(1-x)^2}, f''(0) = -\frac{1}{(1-0)^2} = -1; f'''(x) = -\frac{2}{(1-x)^3}, f'''(0) = -\frac{2}{(1-0)^3} = -2; \\ f^{(4)}(x) &= -\frac{6}{(1-x)^4}, f^{(4)}(0) = -\frac{6}{(1-0)^4} = -6; \end{aligned}$$

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$\begin{aligned} f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots \\ \Rightarrow f(h) &= 0 - h - \frac{h^2}{2!} - \frac{2h^3}{3!} - \frac{6h^4}{4!} + \dots \\ \Rightarrow f(x) &= 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \\ \Rightarrow \ln(1-x) &= 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \\ \Rightarrow \ln(1-x) &= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \end{aligned}$$

(iii) We know, $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$.

So,

$$\begin{aligned}\ln\left(\frac{1+x}{1-x}\right) &= \ln(1+x) - \ln(1-x) \\ &= \left(x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots\right) - \left(-x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots\right) \\ &= 2x + \frac{4x^3}{3!} + \dots\end{aligned}$$

(iv) We know, $\log_a b = \frac{\log_c b}{\log_c a}$.

So,

$$\begin{aligned}\log_{(1-x)}(1+x) &= \frac{\ln(1+x)}{\ln(1-x)} \\ &= \frac{\left(x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots\right)}{\left(-x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots\right)}\end{aligned}$$

Performing algebraic long division, we get,

$$\begin{array}{r} -1 + x - \frac{x^2}{2!} + \frac{5x^3}{12} + \dots \\ -x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \overline{) \quad \quad \quad} \\ \quad \quad \quad x + \frac{x^2}{2!} + \frac{2x^3}{3!} + \frac{6x^4}{4!} + \dots \\ \quad \quad \quad \underline{- \frac{2x^2}{2!} - \frac{12x^4}{4!} + \dots} \\ \quad \quad \quad \quad \quad -x^2 - \frac{x^3}{2!} - \frac{2x^4}{3!} + \dots \\ \quad \quad \quad \quad \quad \underline{\quad \quad \quad \frac{x^3}{2!} - \frac{x^4}{6} + \dots} \\ \quad \quad \quad \quad \quad \quad \quad \frac{x^3}{2!} + \frac{x^4}{2! \times 2!} + \dots \\ \quad \quad \quad \quad \quad \quad \quad \underline{\quad \quad \quad \frac{5x^4}{12} + \dots} \\ \quad \quad \quad \quad \quad \quad \quad \quad - \frac{5x^4}{12} + \dots \\ \quad \quad \quad \quad \quad \quad \quad \quad \quad \vdots \end{array}$$

Hence, we can write,

$$\Rightarrow \log_{(1-x)}(1+x) = -1 + x - \frac{x^2}{2!} + \frac{5x^3}{12} + \dots \quad [\text{Quotient of the above algebraic long division.}]$$

Rubric:

- 6 points for the correct series of $\ln(1+x)$.
- 4 points for the correct series of $\ln(1-x)$.
- 4 points for the correct series of $\ln\left(\frac{1+x}{1-x}\right)$.
- 4 points for the correct series of $\log_{(1-x)}(1+x)$.

3. a) Compare and contrast the following interpolation methods: Direct method, Newton's Divided Difference method, and Lagrangian method.

4
(CO3)
(PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Direct	Newton's Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$	$f_n(x) = b_0 + b_1(x - x_0) + \dots + b_n(x - x_0)(x - x_1) \dots (x - x_{n-1})$	$f_n(x) = \sum_{i=0}^n \left(\prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3 \log(A + b))$, or $O((n + 1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can be optimized further down to $O(n \log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

Rubric:

- 2 points for each valid comparison.

- b) Suppose, there is an unknown function $f(x)$ and you are given 3 points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ through which $f(x)$ passes. Now, to approximate $f(x)$, you decide to use Newton's Divided Difference method of interpolation and obtain a quadratic interpolant $f_2(x)$. Derive the representation of 2^{nd} order Newton's Divided Difference polynomial $f_2(x)$.

9
(CO4)
(PO1)

Solution:



Just click here.

Rubric:

- 1 point for writing the quadratic NDD interpolant $f_2(x)$ in the proper form.
- 1 point for correct derivation of b_0 .
- 2 point for correct derivation of b_1 .
- 4 points for correct derivation of b_2 .
- 1 point for writing the correct expanded form of the quadratic NDD interpolant $f_2(x)$.

- c) The specific heat capacity C of a substance is defined as the amount of heat that is required to raise the temperature of unit mass of that particular substance by 1 degree. Suppose, you are conducting an experiment to determine how much heat is required to bring some volume of water to its boiling point. The values of specific heat capacity C of water that you have

6
(CO2)
(PO1)

calculated at different temperature values T are shown in Table 2.

Table 2: Specific heat C of water as a function of temperature T for Question 3.c

Temperature, T ($^{\circ}\text{C}$)	Specific Heat, C ($\text{Jkg}^{-1}\text{C}^{-1}$)
22	4181
42	4179
52	4186
82	4199
100	4217

Determine the value of the specific heat at $T = 61^{\circ}\text{C}$ using the Direct method of interpolation and a second order polynomial.

Solution:

For second order polynomial interpolation (also called quadratic interpolation), the equation for the specific heat would be —

$$C(T) = a_0 + a_1T + a_2T^2$$

Here, the order $n = 2$. Now, let's choose the nearest $n + 1 = 2 + 1 = 3$ points that bracket the point $T = 61^{\circ}\text{C}$ and evaluate the polynomial. The 3 points are $T_0 = 42$, $T_1 = 52$, and $T_2 = 82$. The respective specific heat values are $C(T_0) = 4179$, $C(T_1) = 4186$, and $C(T_2) = 4199$. Using these values, we obtain the following equations —

$$C(42) = a_0 + a_1 \times 42 + a_2 \times (42)^2 = 4179$$

$$C(52) = a_0 + a_1 \times 52 + a_2 \times (52)^2 = 4186$$

$$C(82) = a_0 + a_1 \times 82 + a_2 \times (82)^2 = 4199$$

Writing these equations in matrix form, we have —

$$\begin{aligned} \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} \frac{533}{50} & -\frac{287}{25} & \frac{91}{50} \\ -\frac{67}{200} & \frac{31}{75} & -\frac{47}{600} \\ \frac{1}{400} & -\frac{1}{300} & \frac{1}{1200} \end{bmatrix} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4135 \\ 1.3267 \\ -6.6667 \times 10^{-3} \end{bmatrix} \end{aligned}$$

Hence,

$$C(T) = 4135 + 1.3267T - 6.6667 \times 10^{-3}T^2; \quad 42 \leq T \leq 82$$

At the point $T = 61$, we get,

$$C(61) = 4135 + 1.3267 \times 61 - 6.6667 \times 10^{-3} \times (61)^2 = 4191.2 \text{ Jkg}^{-1}\text{C}^{-1}$$

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 2 point for the correct coefficient values.
- 2 point for the correct answer.

- d) i. Why is $f_9(x)$ in Figure 3(b) demonstrably worse at capturing the pattern of $f(x)$ compared to $S(x)$ in Figure 3(c)? Explain with necessary illustration(s).

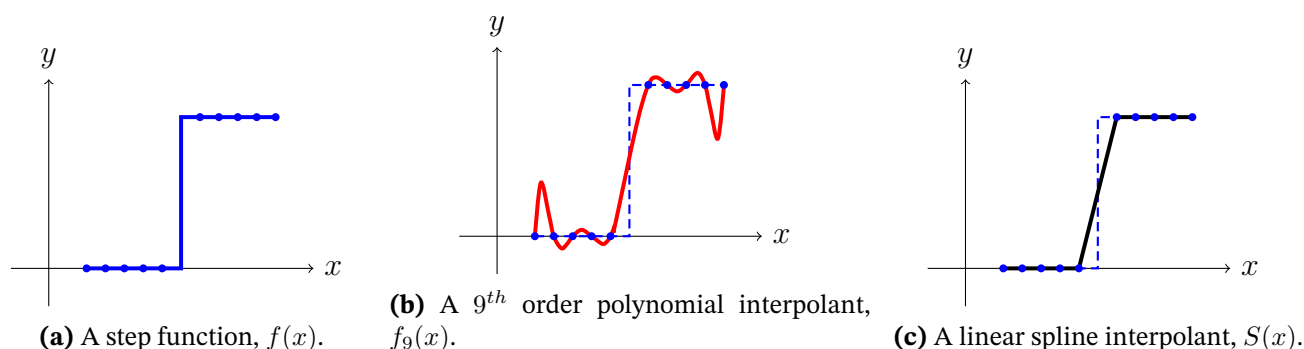


Figure 3: Predicting the nature of $f(x)$ for Question 3.d)i

Solution:

$f_9(x)$ is a worse interpolant than $S(x)$ due to the manifestation of the Runge phenomenon. Let's consider Runge's function $f(x) = \frac{1}{1 + 25x^2}$ to explain the phenomenon. When we try to approximate the nature of this function using a high-order interpolant, we observe an unhinged oscillatory behavior at either end of the interpolant.

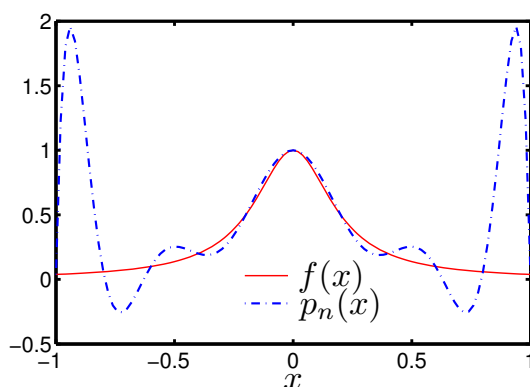


Figure 4: A 9th order polynomial interpolant passing through 10 equidistant points on Runge's function (bad).

However, we can ameliorate this phenomenon to a great extent by using splines.

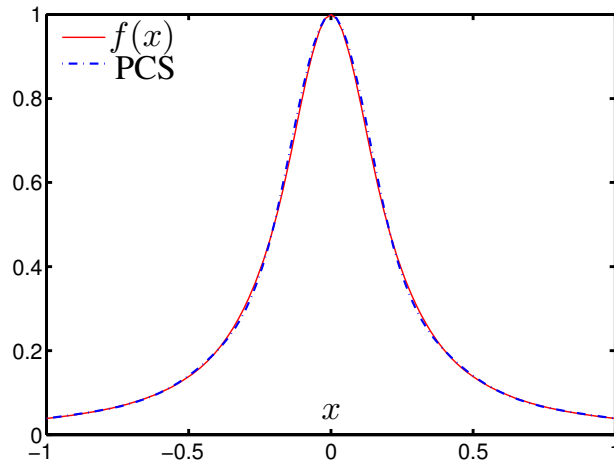


Figure 5: A 9th order cubic spline interpolant passing through 10 equidistant points on Runge's function (good).

As evident from the stark contrast between the interpolants portrayed in Figure 4 and Figure 5, splines are less susceptible to the Runge phenomenon since they stay true to the nature of the underlying function between their respective start and end points.

Rubric:

- 1 point for mentioning the correct reason.
- 3 points for illustrating nice figure(s) that explain(s) the Runge phenomenon.

- ii. Suppose you want to perform Spline Interpolation on $n + 1$ data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\dots$, and $(x_n, f(x_n))$ using Cubic splines of the form $f(x) = ax^3 + bx^2 + cx + d$. How many unknowns and how many equations would you have to deal with? Mention how you would obtain those equations and derive them.

5
(CO4)
(PO1)

Solution:

Each cubic spline has 4 unknowns. So, for n such splines, we have a total of $4n$ unknowns. To obtain these $4n$ unknowns, we need $4n$ equations.

Here, we have $n + 1$ given data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\dots$, and $(x_n, f(x_n))$. The n cubic splines that go through consecutive points and form the interpolant $f(x)$ are,

$$f(x) = \begin{cases} a_1x^3 + b_1x^2 + c_1x + d_1; & \text{for } x_0 \leq x \leq x_1 \\ a_2x^3 + b_2x^2 + c_2x + d_2; & \text{for } x_1 \leq x \leq x_2 \\ \vdots & \vdots \\ a_nx^3 + b_nx^2 + c_nx + d_n; & \text{for } x_{n-1} \leq x \leq x_n \end{cases}$$

Now, let's derive the $4n$ equations. Since the cubic splines pass through consecutive

points, we get the following $2n$ equations.

$$\begin{aligned}
 a_1x_0^3 + b_1x_0^2 + c_1x_0 + d_1 &= f(x_0) \\
 a_1x_1^3 + b_1x_1^2 + c_1x_1 + d_1 &= f(x_1) \\
 &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 a_1x_{i-1}^3 + b_1x_{i-1}^2 + c_1x_{i-1} + d_1 &= f(x_{i-1}) \\
 a_1x_i^3 + b_1x_i^2 + c_1x_i + d_1 &= f(x_i) \\
 &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 a_1x_{n-1}^3 + b_1x_{n-1}^2 + c_1x_{n-1} + d_1 &= f(x_{n-1}) \\
 a_1x_n^3 + b_1x_n^2 + c_1x_n + d_1 &= f(x_n)
 \end{aligned}$$

Now, we need to find out the 1st order derivatives of the consecutive cubic splines at the $n - 1$ interior points, since they are continuous at those points. We obtain another $n - 1$ equations in this way.

$$\begin{aligned}
 3a_1x_1^2 + 2b_1x_1 + c_1 &= 3a_2x_1^2 + 2b_2x_1 + c_2 \\
 &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 3a_{n-1}x_{n-1}^2 + 2b_{n-1}x_{n-1} + c_{n-1} &= 3a_nx_{n-1}^2 + 2b_nx_{n-1} + c_n
 \end{aligned}$$

Again, we need to find out the 2nd order derivatives of the consecutive cubic splines at the $n - 1$ interior points, since they are continuous at those points. We obtain yet another $n - 1$ equations in this way.

$$\begin{aligned}
 6a_1x_1 + 2b_1 &= 6a_2x_1 + 2b_2 \\
 &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 6a_{n-1}x_{n-1} + 2b_{n-1} &= 6a_nx_{n-1} + 2b_n
 \end{aligned}$$

Now, we require 2 more equations. For these, we consider $a_1 = 0$ and $a_{n-1} = 0$. So, in aggregate, we have $2n + (n-1) + (n-1) + 2 = 4n$ equations for the $4n$ unknowns.

Rubric:

- 1 point for the correct number of unknowns and equations.
- 1 point for the initial $2n$ equations.
- 1 point for the $n - 1$ equations using 1st order derivatives.
- 1 point for the $n - 1$ equations using 2nd order derivatives.
- 1 point for the last 2 equations.